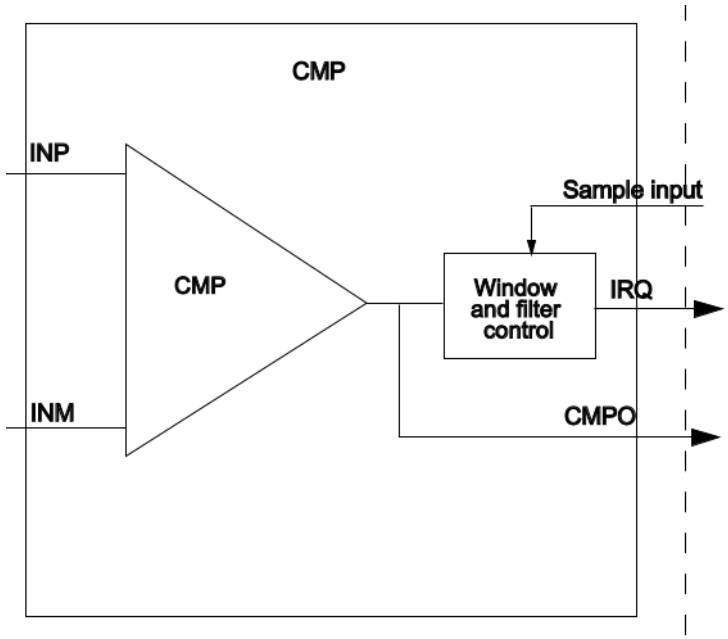
CMP:Comparator



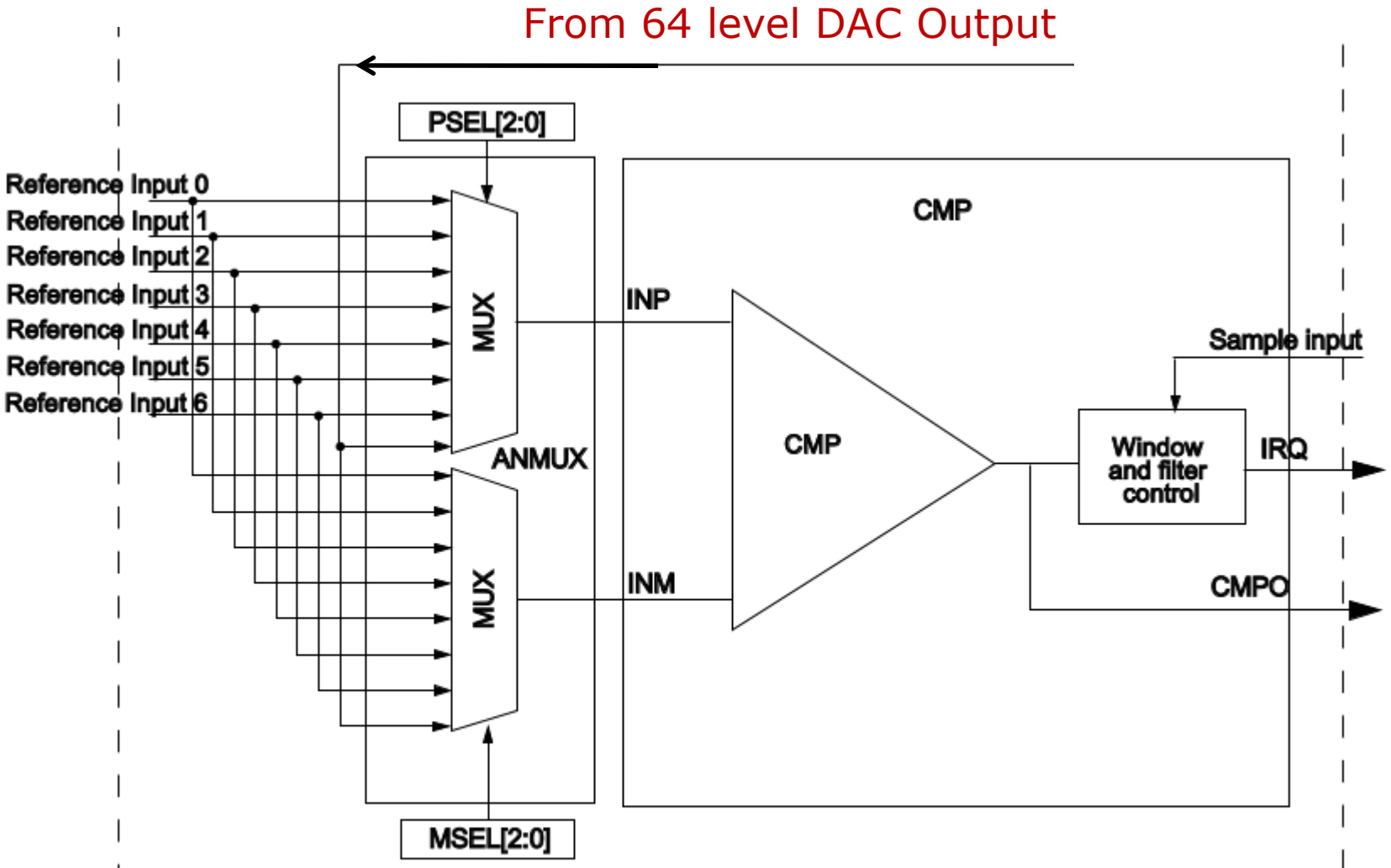
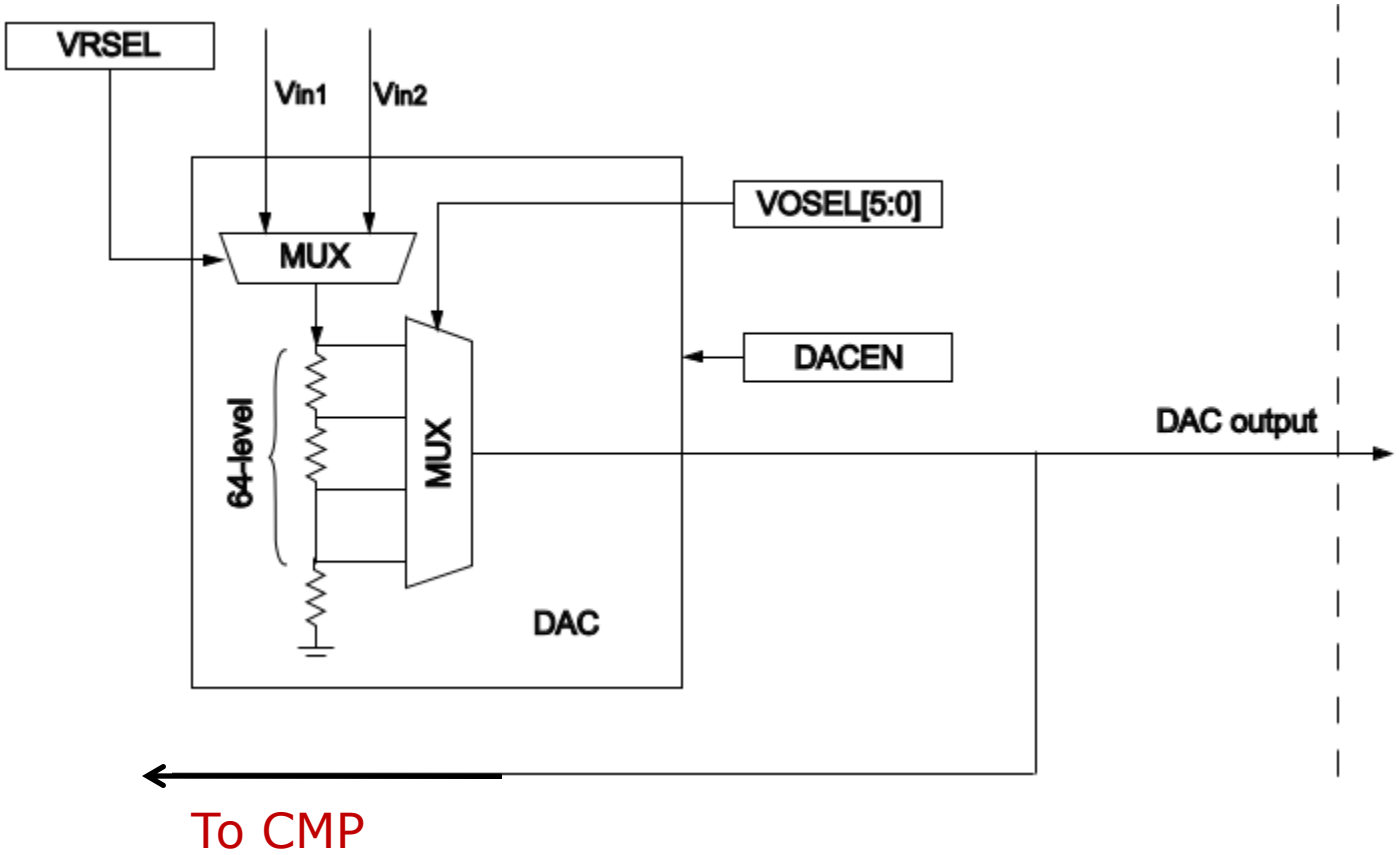


Figure 36-1. CMP, DAC and ANMUX block diagram


VRSEL
Vin1
Vin2
MUX
VOSEL[5:0]
DACEN
64-level
MUX
DAC output
DAC
To CMP

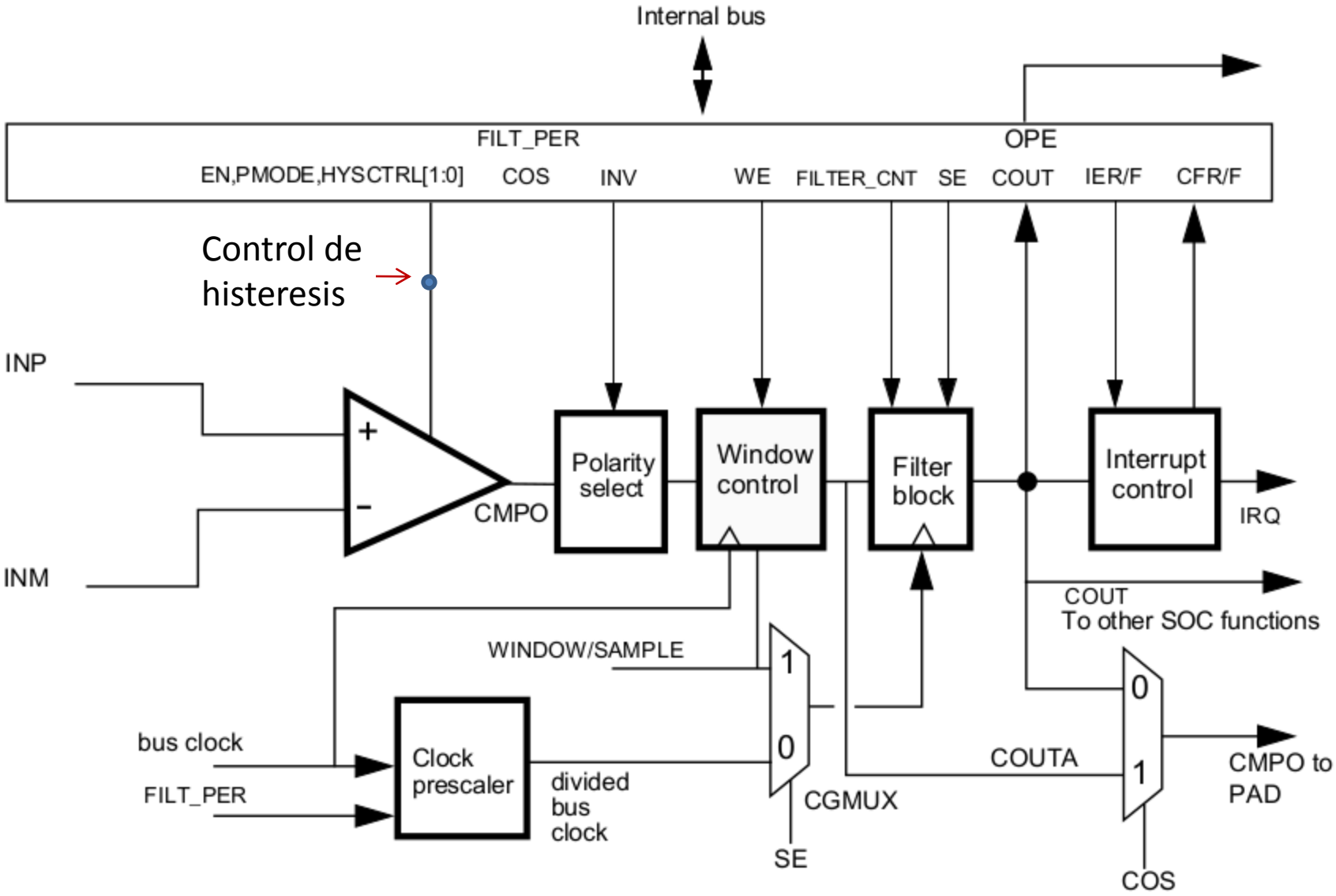


Figure 36-2. Comparator module block diagram

Comparator operation in Continuous mode

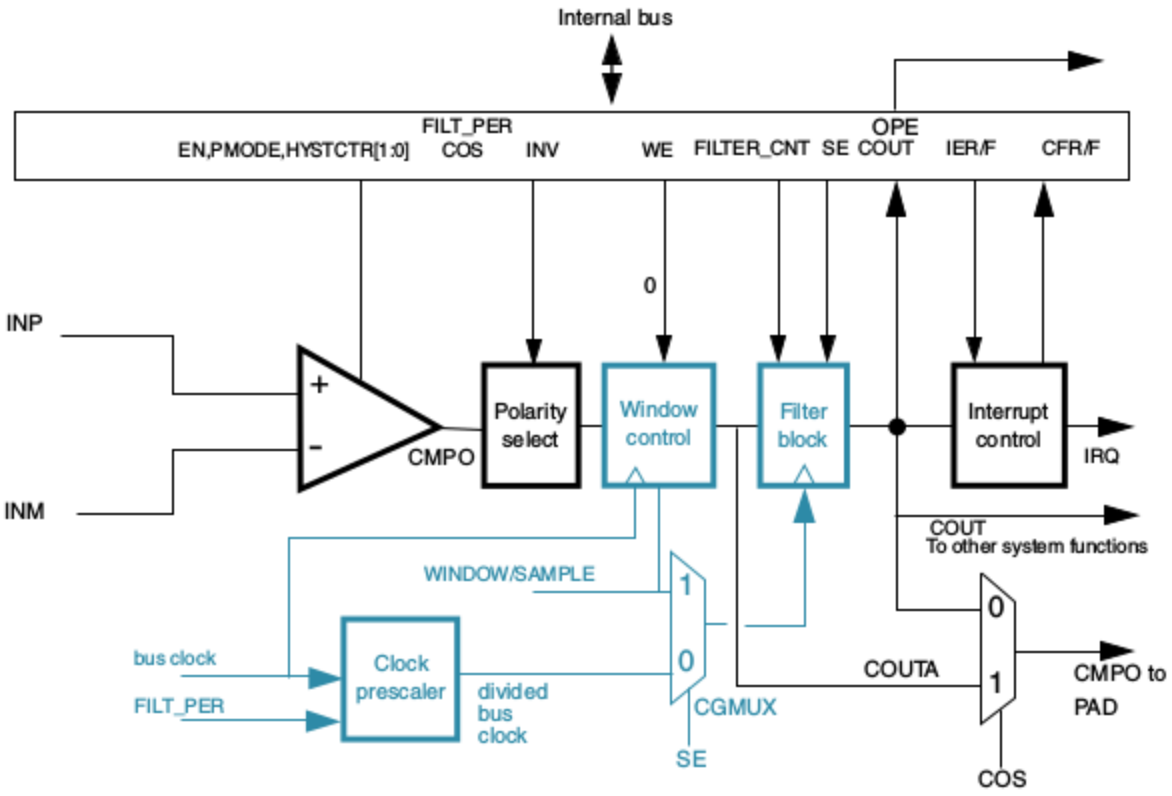


Figure 36-3. Comparator operation in Continuous mode

The path from comparator input pins to output pin is operating in combinational unclocked mode. COUT and COUTA are identical.

Sampled, Non-Filtered mode

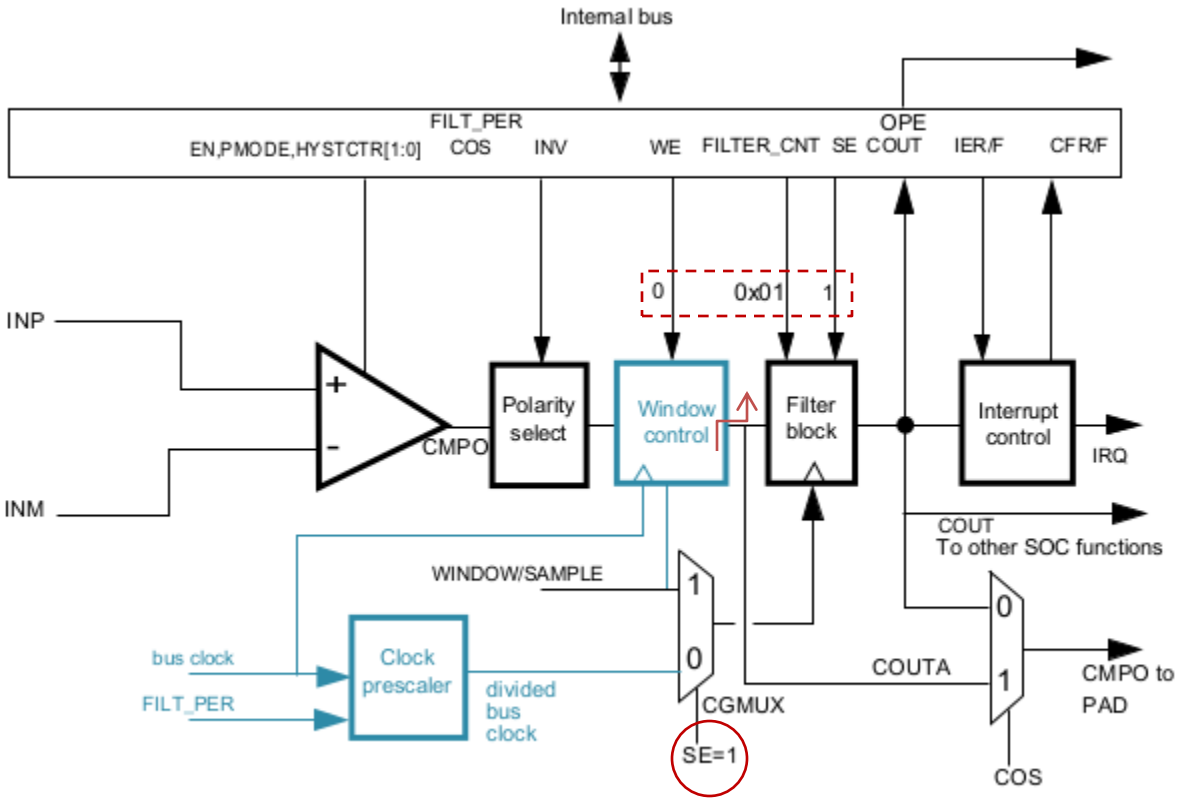


Figure 36-4. Sampled, Non-Filtered (# 3A): sampling point externally driven

COUTA is sampled whenever a rising-edge is detected on the filter block clock input.

Sampled, Non-Filtered mode

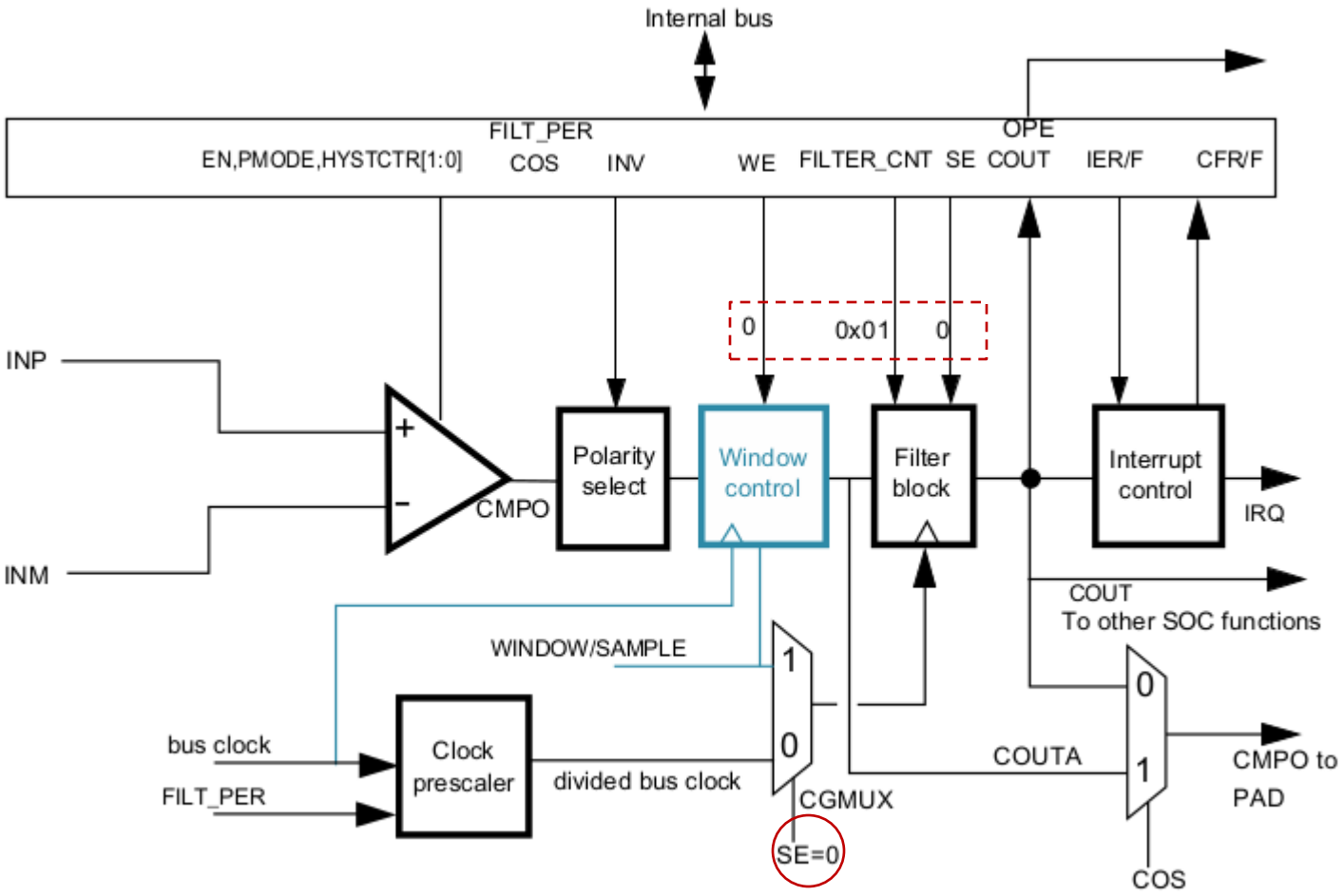


Figure 36-5. Sampled, Non-Filtered (# 3B): sampling interval internally derived

COUTA is sampled whenever a rising-edge is detected on the filter block clock input.

Windowed Mode

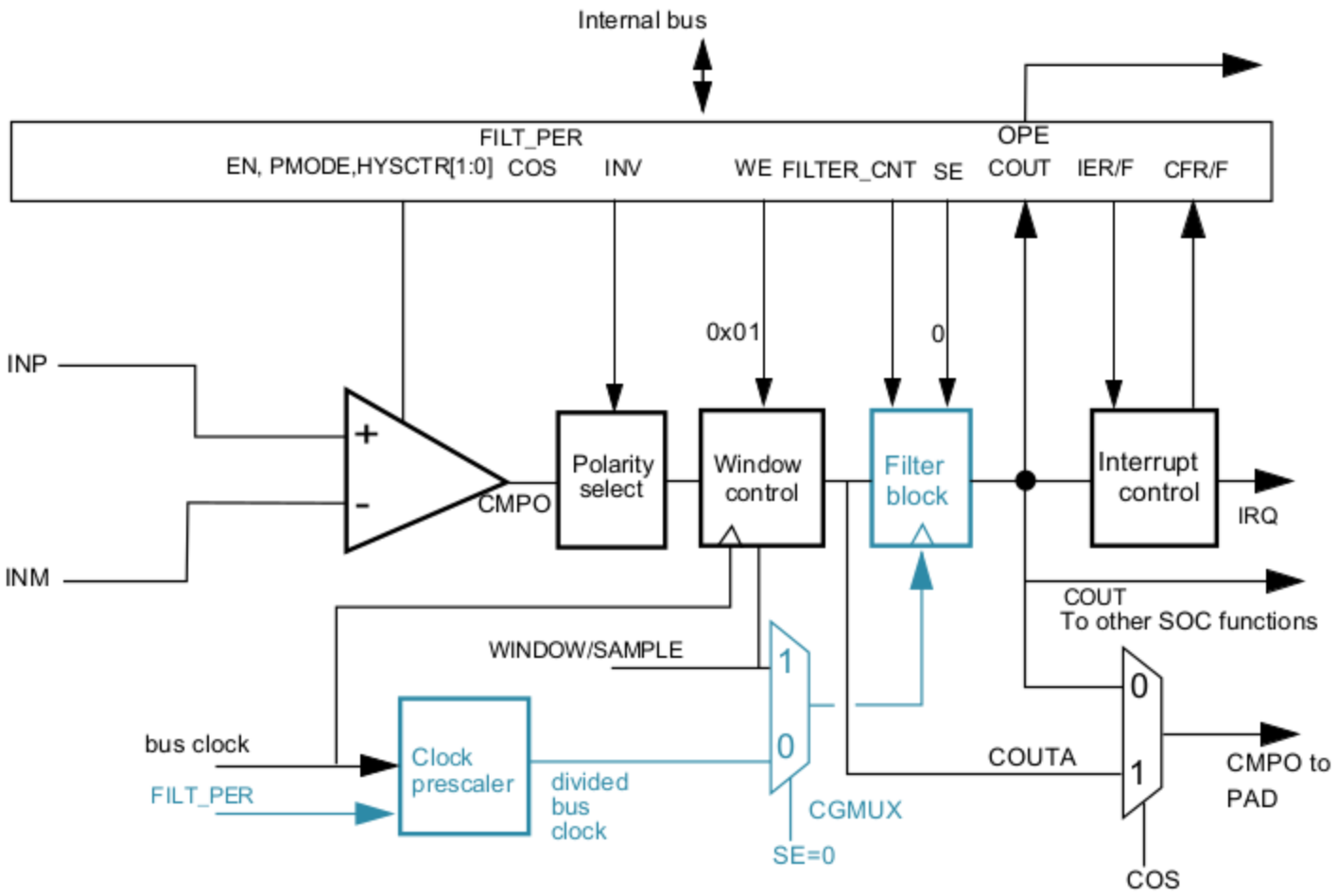


Figure 36-9. Windowed mode

The analog comparator output is passed to COUTA only when the WINDOW signal is high. The last latched value is held when WINDOW = 0.

Windowed Mode

From PDB ←

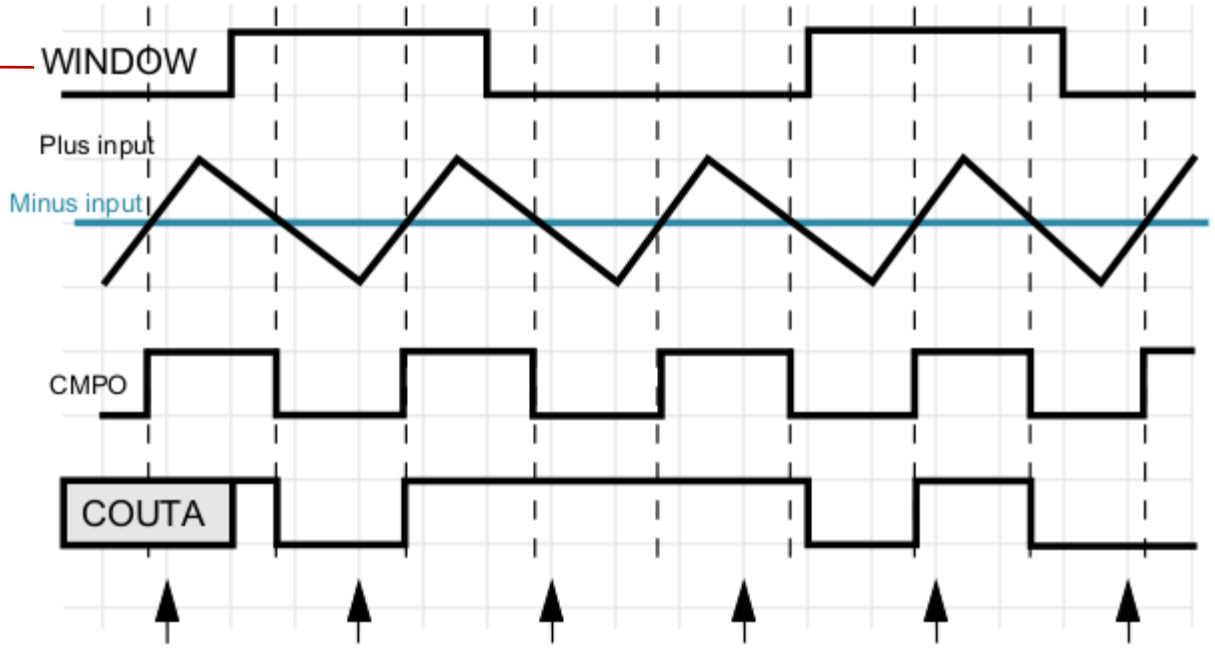


Figure 36-8. Windowed mode operation

The analog comparator output is passed to COUTA only when the WINDOW signal is high. The last latched value is held when WINDOW = 0.